

Reviewer Pack — Minimal Rank of K-Theory Groups over Quantum Entanglement Te...

Entry 4067b7a686d2 · INCONCLUSIVE

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Minimal Rank of K-Theory Groups over Quantum Entanglement Tensor Product

Entry ID: 4067b7a686d2 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 01:38:04 UTC

1. Conjecture

Field A (mathematical branch): Algebraic K-Theory (Groups) **Field B** (complexity object): Complexity Theory: Quantum Entanglement Tensor Product

Statement:

[‘For a read-twice BP P , the minimal rank of its associated K-theory group $\rho_K(P)$ is upper-bounded by $O(\log(2^n))$, where n is the number of variables in P .’, ‘If P is an IP_2 trivial BP, then $\rho_K(P) = \Omega(n)$.’, ‘For all read-twice BPs P , $\rho_K(P) \geq \log(\text{size}(P))$ ’]

Rationale (proposer’s reasoning):

[‘Algebraic K-theory provides a robust framework for studying the structure of rings and modules. Its application to quantum entanglement could reveal new insights into the complexity of computing entangled states.’, ‘The connection between K-theory and quantum entanglement tensor products is novel, as it bridges algebraic topology with quantum information theory.’, ‘This conjecture aims to expose a quantifiable relationship between the structural properties of BPs and the entanglement in their associated quantum systems.’]

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 15835a7c6e9584fd

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the mean of $\rho_K(P)$ over 30 seeds is less than or equal to $O(\log(2^n))$, where n is the number of variables in P , and for all seeds, $\rho_K(P)$ is greater than or equal to $\log(\text{size}(P))$. The conjecture is falsified if any seed produces a ratio of $|\rho_K(P) - \log(\text{size}(P))|$ greater than 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'Algebraic K-Theory Groups' AND 'Quantum Entanglement Tensor Product' AND minimal rank' - 'K-theory group' AND read-twice BP P AND ' $0(\log(2^n))$ ' AND quantum entanglement tensor product - IP_2 trivial BP AND ' $\Omega(n)$ ' AND 'K-theory group' AND 'quantum entanglement tensor product'

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_bp(n):
        bp = []
        for _ in range(2**n):
            row = [random.choice([0, 1]) for _ in range(n)]
            bp.append(row)
        return bp

    def size(bp):
        return len(bp)

    def k_theory_rank(bp):
        n = int(math.log2(len(bp)))
        if n == 0:
            return 0
        rank = 0
        for i in range(n):
            col = [bp[j][i] for j in range(2**n)]
            if sum(col) > 0:
                rank += 1
        return rank

    def is_ip_2_trivial(bp):
        n = int(math.log2(len(bp)))
```

```

    if n == 0:
        return True
    for i in range(n):
        col = [bp[j][i] for j in range(2**n)]
        if sum(col) != 1:
            return False
    return True

def log_size(bp):
    return math.log(size(bp))

n = random.randint(5, 40)
bp = generate_bp(n)
rho_K_P = k_theory_rank(bp)
size_P = size(bp)

if is_ip_2_trivial(bp):
    conjecture_holds = rho_K_P >= n
    counterexample = "IP_2 trivial BP" if not conjecture_holds else ""
else:
    conjecture_holds = rho_K_P <= math.log(2**n) and rho_K_P >= log_size(bp)
    counterexample = "rho_K(P) out of bounds" if not conjecture_holds else ""

return {
    "metric_name": "k_theory_rank",
    "metric_value": rho_K_P,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2**i + 1 for i in range(5, 6)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rho_K_P = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rho_K_P} std=0.0 support_fraction=1.0")
    elif any(abs(r["metric_value"] - log_size(generate_bp(int(math.log2(len(r["instances_tested"])))))) > 0.001 for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if abs(result["metric_value"] - log_size(generate_bp(int(math.log2(len(r["instances_tested"])))))) > 0.001)
        print(f"RESULT: FALSIFIED counterexample='rho_K(P) out of bounds' first_failing_seed={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=insufficient_evidence")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
: {'metric_name': 'K-theory Rank', 'metric_value': 15, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 7, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 25, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 23, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 9, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 38, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 36, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 19, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'K-theory Rank', 'metric_value': 17, 'instances_tested': 1, 'conjecture_holds': False}
RESULT: FALSE
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test has only considered a very small number of instances ($n \leq 15$), which is insufficient to draw conclusions about the conjecture's validity. The metric does not scale trivially with n , but the current evidence is too weak to confirm the conjecture.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results show that for at least one seed, the ratio $|\rho_K(P) - \log(\text{size}(P))|$ exceeds 3, which violates the falsification condition of the conje | next: Further investigation is needed to determine if there are any patterns or conditions under which the conjecture holds. Additional testing with a larger number of seeds and different types of read-twice BPs is recommended.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11345
2	propose	ollama_remote	glm4:latest	0	0	10207
3	preregistration	ollama_remote	glm4:latest	0	0	6234
4	novelty	ollama_remote	glm4:latest	0	0	5086
5	novelty	ollama_remote	glm4:latest	0	0	5302
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14823
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7708
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9741
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8797
10	critic	ollama_remote	glm4:latest	0	0	11286
11	judge	ollama_remote	glm4:latest	0	0	6328

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 96858 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in [research/audit/4067b7a686d2.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/4067b7a686d2.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/4067b7a686d2.tar.gz` (if generated)